

राष्ट्रीय प्रौद्योगिकी संस्थान, हमीरपुर (हि.प्र.)
National Institute of Technology Hamirpur (H.P.)
(Under Ministry of Education, Govt. of India, New Delhi)

Branch/ Semester	: B.Tech. (CSE)	Subject Code	: CS-322
Subject Name	: Software Engineering	Duration	: 3.00 Hrs
Date	: 01/05/2025 (09:00 to 12:00)	Max. Marks	: 50

CO1: Understand and analyze the concept of software development and software engineering.
CO2: Compare and comprehend different software engineering process models.
CO3: Design of software projects and do the cost estimation.
CO4: Apply different software testing techniques

1. (a) How do standards support interoperability and consistency in software projects?
(CO3) [3] -2
- (b) Explain in detail the unified process model. Compare it with iterative enhancement model.
(CO1) [4] 6
- (c) What do you understand by total quality management (TQM)? What are the advantages of TQM? Does ISO 9000 standard aim for TQM?
(CO4) [3] =
2. (a) Discuss the key quality attributes of software. How do they affect the overall software engineering process?
(CO4) [2] =
- (b) Distinguish between software verification and software validation. Can one be used in place of the other? Justify your answer?
(CO4) [3] 1.5
- (c) Draw control flow graph and calculate cyclomatic complexity for the given code.

```
1. { int i, j, k;  
2.   for (i=0 ; i<=N ; i++)  
3.     p[i] = 1;  
4.   for (i=2 ; i<=N ; i++)  
5.   {  
6.     k = p[i]; j=1;  
7.     while (a[p[j]-1] > a[k] {  
8.       p[j] = p[j-1];  
9.       j--;  
10.    }  
11.    p[j]=k;  
12. }
```


(CO4) [5] 3
3. (a) How do black-box and white-box testing techniques differ in terms of their approach and application?
(CO4) [3] 2.6
- (b) What are the basic principles of programming during the coding phase? How does structured programming help improve code readability
(CO4) [4] -2
- (c) Compare and contrast function-oriented design and object-oriented design. Discuss the advantages and limitations of both approaches.
(CO3) [3] -2

Roll Number: _____

4. (a) Define the following:

- i) Actors ii) Scenario
v) Sequence Diagram

iii) Use cases

iv) Class Diagram

(CO3) [5]

(b) What are the key architecture styles for component and connector views? Discuss their significance in system design? (CO3) [3]

(c) Is it true that whenever you increase the cohesion of your design, coupling in the design would automatically decrease? Justify your answer by using suitable examples. (CO3) [2]

5. (a) A project size of 200 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the effort, development time, average staff size and productivity of the project. (CO3) [4]

Software Project	a_0	b_0	c_0	d_0
Organic	2.4	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

(b) Compare the advantages and disadvantages of the following two project size estimation techniques—expert judgement and Delphi technique? (CO2) [3]

(c) Define the software development process. What are the key characteristics of a good software process? (CO1) [3]



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Branch/ Semester	: CSE/6 th	Subject Code	: CS-322
Subject Name	: Software Engineering	Duration	: 1.5 Hrs
Date	: 25/02/2025	Max. Marks	: 30

- Q.1 What do you understand by the term software development life cycle? Why is it necessary to model software life cycle and to document it? What problems might a software development organisation face if it is not following any SDLC for development of a large-sized software? 6
- Q. 2 Why is the classical waterfall model called an idealistic development model? Does this model of development has any practical use at all? 6
- Q.3 Both the prototyping model as well as the spiral model have been designed to handle risks. Identify how exactly risk is handled in each. How do these two models can be compared with respect to their risk handling capabilities? 6
- Q.4 How software should be chosen? Explain the complete process. 6
- Q.5 What were the challenges in software development process? Discuss. 6